



The SK Telecom Breach:

A Four-Year Undetected APT Attack on Korea's Largest Telecom Network

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What is SK Telecom(SKT)?

**The largest mobile carrier
in South Korea**

like AT&T / Verizon



27 million subscribers

Korea's population: around 52 million

**Operator of nationwide
5G network, IoT infrastructure,
digital ID services**



**Considered one of the most
security-advanced telecom
companies in Korea**

Incident Overview

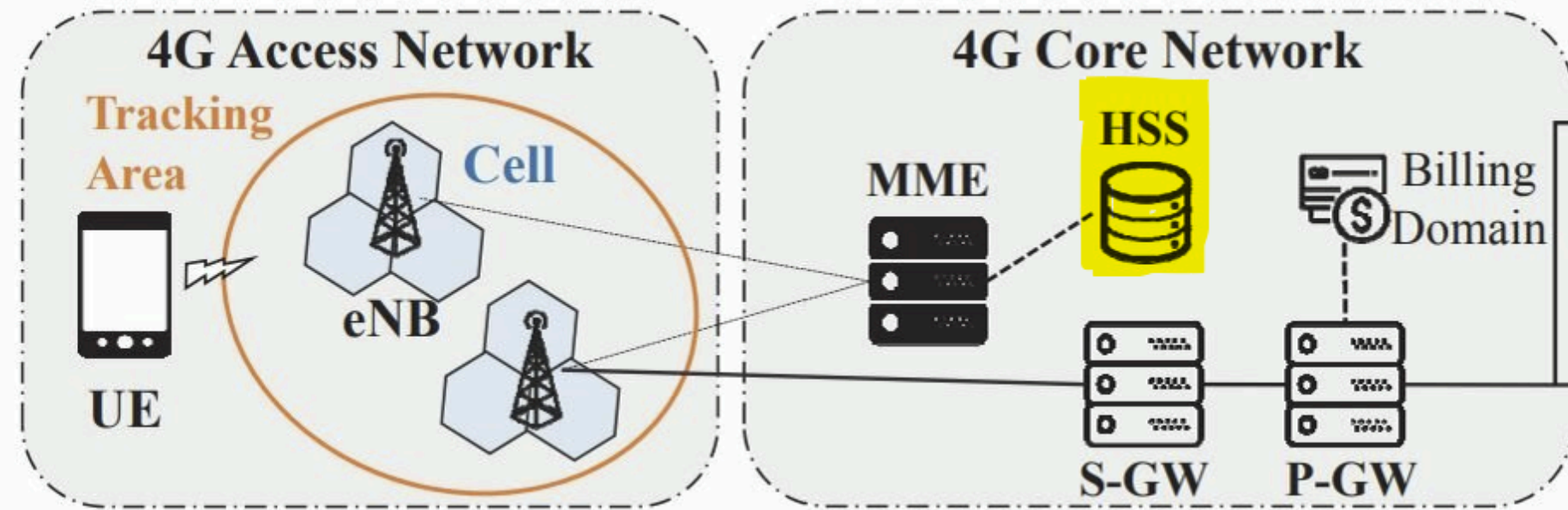
- **Date disclosed** : April 18, 2025
- **Core system breached:** HSS subscriber authentication server (3 servers)
- **Data leaked** : 9.82GB (IMSI, phone numbers, USIM keys, email, etc.)
- **Risk level** : Enables SIM cloning, 2FA interception, identity theft
- **Why this matters** : SIM data leak $\approx$ digital identity compromised

IMSI* : International Mobile Subscriber Identity.

Globally unique number that identifies a mobile subscriber on cellular networks

Technical Analysis — How the Attack Happened

1) HSS



<Structure of the 4G mobile network>

- Core database storing subscriber identity & authentication data
- Contains IMSI, SIM authentication key (Ki), phone numbers, service profiles
- Compromise = attacker gains “master key” to mobile identity system

2) Attack Method: APT Intrusion & BPFDoor

APT(Advanced Persistent Threat) → Attack strategy



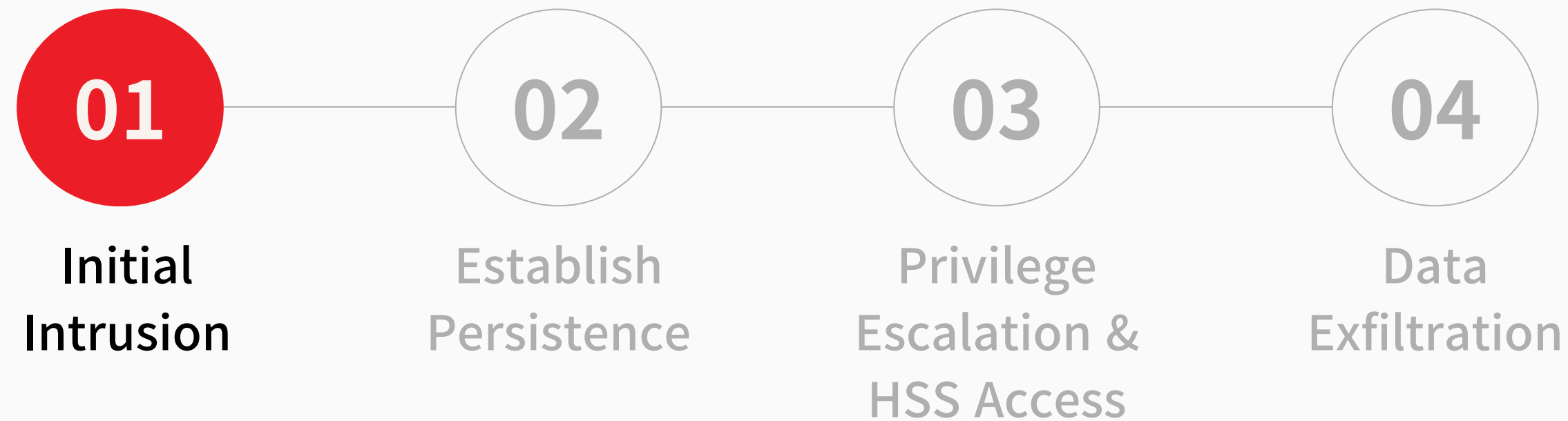
→ reach high-value systems

BPFDoor → Tool / Method

- Stealth Linux Backdoor Malware
- Port-less backdoor (no listening ports → very hard to detect)
- Receives remote attacker commands covertly
- Enables persistence, system control, data extraction

Technical Analysis — How the Attack Happened

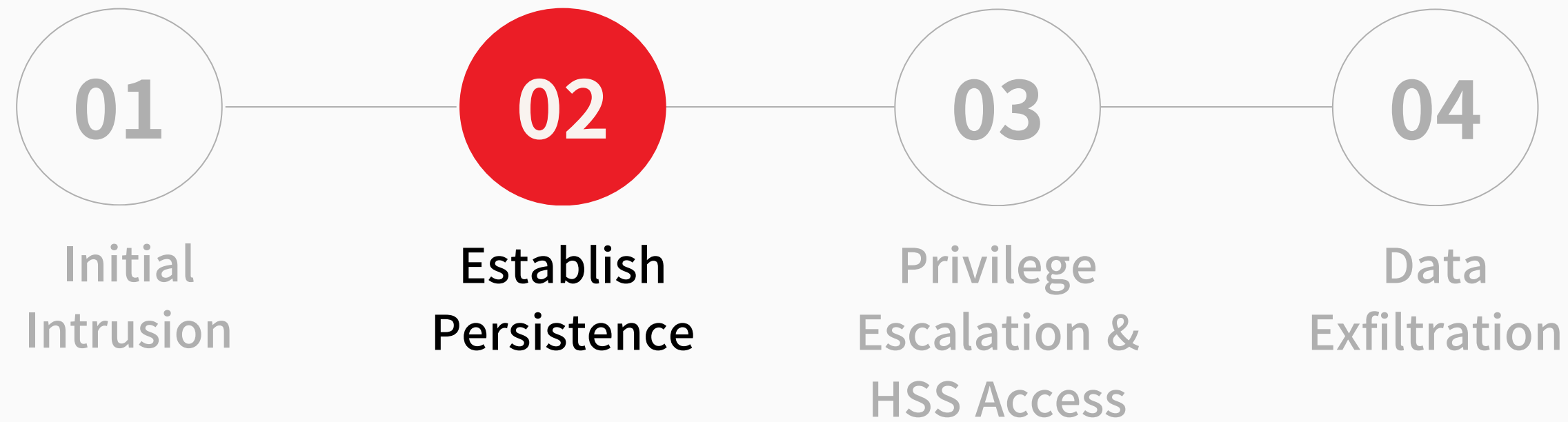
3) 4-Step Attack Chain



Initial Intrusion

Initial intrusion via VPN and exposed-system vulnerabilities

3) 4-Step Attack Chain



Establish Persistence

BPFDoor installed → long-term stealth presence

- Port-less Linux backdoor; extremely hard to detect
- Allowed lateral movement inside internal network

3) 4-Step Attack Chain



Privilege Escalation & HSS Access

Gained admin credentials and reached HSS servers

- Stole admin accounts
- Accessed core subscriber authentication systems (HSS)

Technical Analysis — How the Attack Happened

3) 4-Step Attack Chain



Data Exfiltration

Extracted ~9.7GB of subscriber identity data

- IMSI, Ki, phone numbers, subscriber profiles
- Large outbound traffic finally triggered detection

Impact

Financial Impact

- Stock price dropped 8.5% ($\approx$ \$552 million in market value) $\rightarrow$ fell behind competitor in market valuation
- Regulatory fine: $\approx$ \$92.8 million
- Operating profit fell 90.9% YoY (\$340 million $\rightarrow$ \$33.4 million)

Customer Impact

- About 430,000 users left SKT after the incident
- Significant trust damage to telecom-based authentication

National Impact

- SIM data leak $\approx$ compromise of national digital identity system
- Triggered nationwide security audits & regulatory reforms

Response & Failures

Early Warning Ignored (2021–2022)

- Attack began in 2021; infected servers were found in 2022
- SKT took no significant action → attacker remained undetected for years

Weak Monitoring of HSS (Critical System)

- HSS updates/patches deferred due to “system stability concerns”
- Result: inadequate security monitoring around the most critical system

Legacy Architecture Issues

- HSS consisted of multiple independent legacy databases
- No strict access separation → compromise of one node exposed all nodes

Insufficient Logging (2-year blind spot)

- Memory logging was severely lacking
- Almost no records of attacker activity from initial entry to HSS access

Reference

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Thank you!

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